



A 112-Bit Binary Rescue Payload for BeiDou-3 Short Messaging

Environment-Stratified Reliability, Latency, and Autonomous UAV Integration

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1 · Introduction

When a disaster destroys terrestrial networks, survivors cannot call for help and rescuers cannot locate them — yet a survivor's position is often already known to centimetre accuracy. BeiDou-3 Short Message Communication carries short data

2 · Objectives

- Capacity — can a complete six-field rescue record fit one regional short message?
- Reliability — does delivery survive real propagation environments?

3 · Methods

A survivor's RTK-corrected coordinate is encoded into a fixed-point 112-bit binary payload — latitude and longitude at 7 dp (~cm), altitude, uncertainty radius, triage priority, and survivor ID — transmitted through the BeiDou regional service as an ESP32 + BDS-3 RDSS node, read from the operator portal, decoded, and passed to the ROS 2 mission-planning /target/emergency_coordinate ROS 2 topic that mission-planning and ground-control modules consume. A 232-transmission field campaign across four environments measured delivery reliability is reported with Wilson score intervals and a chi-squared homogeneity test, and every figure regenerates from committed scripts.

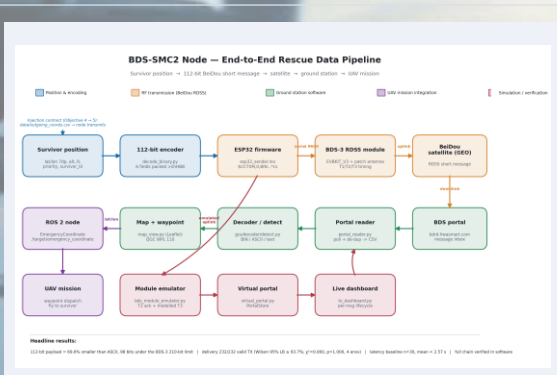


Figure 1. End-to-end BeiDou rescue pipeline: survivor position → 112-bit encode → BeiDou regional service → portal → decode → ROS 2 mission trigger.

4 · Results

Across 232 valid transmissions the satellite segment acknowledged every one (Wilson 95% lower bound $\geq 98.4\%$ pooled, $\geq 93.7\%$ per environment; the rule of three bounds pooled

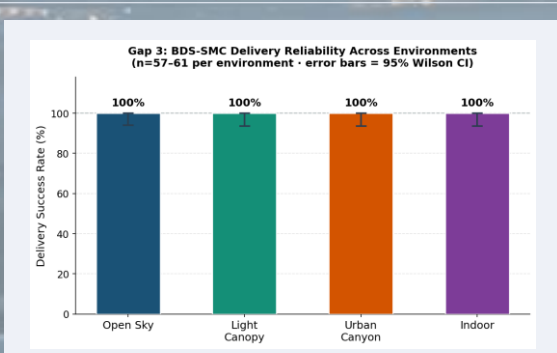


Figure 2. BeiDou delivery reliability by environment with Wilson 95% intervals (n = 57–61 per environment).

5 · Discussion

Among infrastructure-free links, only satellite systems survive a disaster: COSPAS-SARSAT carries no user-definable triage payload and

6 · Conclusions

A complete survivor record fits one 112-bit BeiDou message and decodes losslessly. The link delivered 232/232 transmissions (Wilson $\geq 93.7\%$).

7 · Future Work

- Physical satellite transmission of the 112-bit frame (the one open boundary)
- Diurnal latency characterisation on the operational device

References

- [1] China Satellite Navigation Office. BeiDou-3 RDSS / Short Message Service ICD.
- [2] COSPAS-SARSAT. T.001 — 406 MHz Distress Beacon Specification.

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HEADLINE RESULTS

232 / 232

satellite-acknowledged transmissions, 4 environments

$\geq 93.7\%$

Wilson 95% lower bound per env ($\geq 98.4\%$ pooled)

2.57 s

mean end-to-end latency · n = 30 · all < 5 s

112-bit

complete six-field payload · ~57.6% vs ASCII

bit-exact

decode vs ground truth · fully reproducible

Payload Efficiency

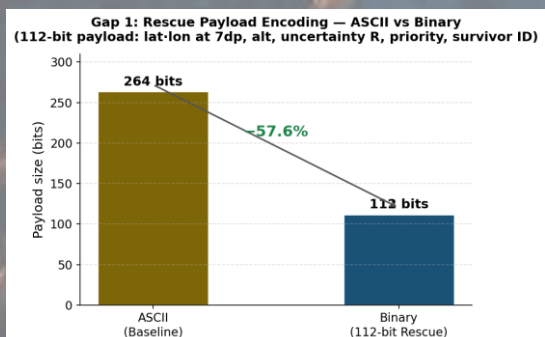


Figure 3. 264-bit ASCII rescue record packs to 112 bits (~57.6%); bit-exact round-trip.

Latency Distribution

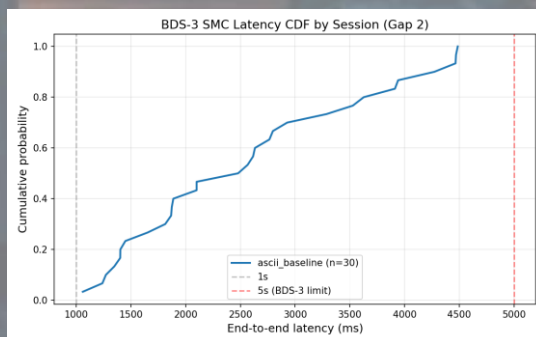


Figure 4. End-to-end latency CDF; every message under 5 s (n = 30).